2.39 1st Order DEs: Integrating Factors

Worked Solutions

Note for Integrating Factors:

We can take the integrating factor to be any non-zero constant multiple of an integrating factor. For example,

$$e^{\int \frac{1}{x}\mathrm{d}x} = e^{\ln|x|} = |x|$$

On any interval not containing $x = 0$, $|x|$ differs from x only by a constant factor (namely ± 1), so we may take the integrating factor to be x. So throughout these worked solutions, the modulus signs will be dropped when multiplying the differential equation through by the integrating factor.

1 a)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(6xy) &= 6x \times \frac{\mathrm{d}y}{\mathrm{d}x} + 6 \times y \\ &= 6x\frac{\mathrm{d}y}{\mathrm{d}x} + 6y\end{aligned}$$

1 b)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(e^{3x}y) &= e^{3x} \times \frac{\mathrm{d}y}{\mathrm{d}x} + 3e^{3x} \times y \\ &= e^{3x}\frac{\mathrm{d}y}{\mathrm{d}x} + 3e^{3x}y\end{aligned}$$

1 c)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(y\tan 4x) &= y \times 4\sec^2 4x + \frac{\mathrm{d}y}{\mathrm{d}x} \times \tan 4x \\ &= \tan 4x\frac{\mathrm{d}y}{\mathrm{d}x} + 4y\sec^2 4x\end{aligned}$$

1 d)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(y\ln 2x) &= y \times \frac{1}{x} + \frac{\mathrm{d}y}{\mathrm{d}x} \times \ln 2x \\ &= \ln 2x\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x}\end{aligned}$$

2 a)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(yx^2) &= y \times 2x + \frac{\mathrm{d}y}{\mathrm{d}x} \times x^2 \\ &= x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy\end{aligned}$$

2 b)
i)

$$\begin{aligned} x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy &= 4 \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(yx^2) &= 4 \\ \Rightarrow yx^2 &= 4x + c \\ \Rightarrow y &= \frac{4x+c}{x^2} \end{aligned}$$

ii)

$$\begin{aligned} x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy &= 4x \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(yx^2) &= 4x \\ \Rightarrow yx^2 &= 2x^2 + c \\ \Rightarrow y &= \frac{2x^2+c}{x^2} \end{aligned}$$

3 a)

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}x}(y\sin x) &= y \times \cos x + \frac{\mathrm{d}y}{\mathrm{d}x} \times \sin x \\ &= \sin x\frac{\mathrm{d}y}{\mathrm{d}x} + y\cos x \end{aligned}$$

3 b)
i)

$$\begin{aligned} \sin x\frac{\mathrm{d}y}{\mathrm{d}x} + y\cos x &= 3x^2 \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\sin x) &= 3x^2 \\ \Rightarrow y\sin x &= x^3 + c \\ \Rightarrow y &= \frac{x^3+c}{\sin x} \end{aligned}$$

ii)

$$\begin{aligned} \sin x\frac{\mathrm{d}y}{\mathrm{d}x} + y\cos x &= e^x \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\sin x) &= e^x \\ \Rightarrow y\sin x &= e^x + c \\ \Rightarrow y &= \frac{e^x+c}{\sin x} \end{aligned}$$

4 a)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(e^{5x}y) &= e^{5x} \times \frac{\mathrm{d}y}{\mathrm{d}x} + 5e^{5x} \times y \\ &= e^{5x}\frac{\mathrm{d}y}{\mathrm{d}x} + 5e^{5x}y\end{aligned}$$

4 b)

$$\begin{aligned}e^{5x}\frac{\mathrm{d}y}{\mathrm{d}x} + 5e^{5x}y &= 6e^{3x} \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{5x}y) &= 6e^{3x} \\ \Rightarrow e^{5x}y &= 2e^{3x} + c \\ \Rightarrow y &= \frac{2e^{3x} + c}{e^{5x}}\end{aligned}$$

4 c)

$$\begin{aligned}0 &= \frac{2e^{3\ln 2} + c}{e^{5\ln 2}} \\ \Rightarrow 0 &= \frac{2e^{\ln 2^3} + c}{e^{\ln 2^5}} \\ \Rightarrow 0 &= \frac{2e^{\ln 8} + c}{e^{\ln 32}} \\ \Rightarrow 0 &= \frac{2 \times 8 + c}{32} \\ \Rightarrow 0 &= 16 + c \\ \Rightarrow c &= -16 \\ \therefore y &= \frac{2e^{3x} - 16}{e^{5x}}\end{aligned}$$

5 a)

$$\begin{aligned}x^4\frac{\mathrm{d}y}{\mathrm{d}x} + 4x^3y &= x^8 \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^4y) &= x^8 \\ \Rightarrow x^4y &= \frac{1}{9}x^9 + c \\ \Rightarrow y &= \frac{1}{9}x^5 + cx^{-4}\end{aligned}$$

5 b)

$$\begin{aligned}
\cos 2x\frac{\mathrm{d}y}{\mathrm{d}x} - 2y\sin 2x &= \cos 2x \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\cos 2x) &= \cos 2x \\
\Rightarrow y\cos 2x &= \frac{1}{2}\sin 2x + c \\
\Rightarrow y &= \frac{\sin 2x}{2\cos 2x} + \frac{c}{\cos 2x} \\
\Rightarrow y &= \frac{1}{2}\tan 2x + c\sec 2x
\end{aligned}$$

5 c)

$$\begin{aligned}
\tan 3x\frac{\mathrm{d}y}{\mathrm{d}x} + 3y\sec^2 3x &= \cos 3x \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\tan 3x) &= \cos 3x \\
\Rightarrow y\tan 3x &= \frac{1}{3}\sin 3x + c \\
\Rightarrow y &= \frac{\sin 3x}{3\tan 3x} + \frac{c}{\tan 3x} \\
\Rightarrow y &= \frac{1}{3}\sin 3x \times \frac{\cos 3x}{\sin 3x} + c\cot 3x \\
\Rightarrow y &= \frac{1}{3}\cos 3x + c\cot 3x
\end{aligned}$$

5 d)

$$\begin{aligned}
e^{-9x}\frac{\mathrm{d}y}{\mathrm{d}x} - 9e^{-9x}y &= e^{-8x} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{-9x}y) &= e^{-8x} \\
\Rightarrow e^{-9x}y &= -\frac{1}{8}e^{-8x} + c \\
\Rightarrow y &= -\frac{1}{8}e^{x} + ce^{9x}
\end{aligned}$$

6 a)

$$\begin{aligned}
&e^{\int 2x\mathrm{d}x} \\
&= e^{x^2}
\end{aligned}$$

6 b)

$$\begin{aligned}
&e^{\int \frac{1}{x}\mathrm{d}x} \\
&= e^{\ln|x|} \\
&= |x|
\end{aligned}$$

6 c)

$$
\begin{aligned}
& e^{\int \frac{1}{2x+1}\mathrm{d}x} \\
&= e^{\frac{1}{2}\ln|2x+1|} \\
&= e^{\ln|(2x+1)^{\frac{1}{2}}|} \\
&= |(2x+1)^{\frac{1}{2}}| \\
&= (2x+1)^{\frac{1}{2}}
\end{aligned}
$$

6 d)

$$
\begin{aligned}
& e^{\int \tan x \mathrm{d}x} \\
& e^{\int \frac{\sin x}{\cos x}\mathrm{d}x} \\
&= e^{-\ln|\cos x|} \\
&= e^{\ln|(\cos x)^{-1}|} \\
&= e^{\ln|\sec x|} \\
&= |\sec x|
\end{aligned}
$$

7 a)
Integrating Factor

$$
\begin{aligned}
&= e^{\int 8\mathrm{d}x} \\
&= e^{8x}
\end{aligned}
$$

7 b)
Integrating Factor

$$
\begin{aligned}
&= e^{\int(-3)\mathrm{d}x} \\
&= e^{-3x}
\end{aligned}
$$

7 c)
Integrating Factor

$$
\begin{aligned}
&= e^{\int(-x)\mathrm{d}x} \\
&= e^{-\frac{1}{2}x^2}
\end{aligned}
$$

7 d)
Integrating Factor

$$
\begin{aligned}
&= e^{\int(2x+1)\mathrm{d}x} \\
&= e^{x^2+x}
\end{aligned}
$$

8 a)

$$x^3\frac{\mathrm{d}y}{\mathrm{d}x} + x^3y = \frac{2}{e^x}$$
$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + y = \frac{2}{x^3e^x}$$

Integrating Factor

$$= e^{\int 1\mathrm{d}y}$$
$$= e^x$$

8 b)

$$\frac{\mathrm{d}y}{\mathrm{d}x} \times e^x + y \times e^x = \frac{2}{x^3e^x} \times e^x$$
$$\Rightarrow e^x\frac{\mathrm{d}y}{\mathrm{d}x} + ye^x = \frac{2}{x^3}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(ye^x) = 2x^{-3}$$
$$\Rightarrow ye^x = -x^{-2} + C$$
$$\Rightarrow y = -\frac{1}{x^2e^x} + \frac{C}{e^x}$$

8 c)
i)

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} \times 2e^x + y \times 2e^x &= \frac{2}{x^3 e^x} \times 2e^x \\
\Rightarrow 2e^x \frac{\mathrm{d}y}{\mathrm{d}x} + 2ye^x &= \frac{4}{x^3} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(2ye^x) &= 4x^{-3} \\
\Rightarrow 2ye^x &= -2x^{-2} + c_1 \\
\Rightarrow y &= -\frac{2}{x^2 \times 2e^x} + \frac{c_1}{2e^x} \\
\Rightarrow y &= -\frac{1}{x^2 e^x} + \frac{c_1}{2e^x}
\end{aligned}$$

This is the same as the answer in b), where $\frac{c_1}{2} = C$
ii)

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} \times e^{x+2} + y \times e^{x+2} &= \frac{2}{x^3 e^x} \times e^{x+2} \\
\Rightarrow e^{x+2} \frac{\mathrm{d}y}{\mathrm{d}x} + ye^{x+2} &= \frac{2e^{x+2}}{x^3 e^x} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(ye^{x+2}) &= \frac{2e^x \times e^2}{x^3 e^x} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(ye^{x+2}) &= 2e^2 x^{-3} \\
\Rightarrow ye^{x+2} &= -e^2 x^{-2} + c_2 \\
\Rightarrow ye^{x+2} &= -\frac{e^2}{x^2} + c_2 \\
\Rightarrow y &= -\frac{e^2}{x^2 e^{x+2}} + \frac{c_2}{e^{x+2}} \\
\Rightarrow y &= -\frac{e^2}{x^2 e^x \times e^2} + \frac{c_2}{e^x \times e^2} \\
\Rightarrow y &= -\frac{1}{x^2 e^x} + \frac{c_2}{e^x \times e^2}
\end{aligned}$$

This is the same as the answer in b), where $\frac{c_2}{e^2} = C$

9 a)

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mu(x)y) = \mu(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mu\prime(x)y$$

For the left hand side of the equation to equal the right hand side of the equation, we must have:

$$\begin{aligned}
\mu(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mu(x)\mathrm{P}(x)y &= \mu(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mu\prime(x)y \\
\Rightarrow \mu(x)\mathrm{P}(x)y &= \mu\prime(x)y \\
\Rightarrow \mu\prime(x) &= \mathrm{P}(x)\mu(x) \quad \text{as required}
\end{aligned}$$

9 b)

$$\begin{aligned} \mu\prime(x) &= \mathrm{P}(x)\mu(x) \\ \Rightarrow \frac{\mu\prime(x)}{\mu(x)} &= \mathrm{P}(x) \end{aligned}$$

Integrating both sides with respect to x:

$$\begin{aligned} \Rightarrow \int \frac{\mu\prime(x)}{\mu(x)}\mathrm{d}x &= \int \mathrm{P}(x)\mathrm{d}x \\ \Rightarrow \ln|\mu(x)| &= \int \mathrm{P}(x)\mathrm{d}x \\ \Rightarrow \mu(x) &= e^{\int \mathrm{P}(x)\mathrm{d}x} \end{aligned}$$

10 a)
When calculating the integrating factor, Archie should instead have got:

$$e^{\int -3\mathrm{d}x} = e^{-3x}$$

10 b)
The integrating factor $\mu(x)$ must satisfy $\mu\prime(x) = \mathrm{P}(x)\mu(x)$
In this example $\mu(x) = -3e^{-3x}$ and $\mathrm{P}(x) = -3$

$$\therefore 9e^{-3x} = -3 \times -3e^{-3x}$$

So Archie's integrating factor $-3e^{-3x}$ does satisfy the required condition, and is therefore a valid integrating factor, even though it is incorrect based on his workings.

10 c)
Archie's solution is $y = e^{3x}\left(x - \frac{c}{3}\right)$
The other student's solution is $y = e^{3x}(x + c)$
If we let $A = -\frac{c}{3}$ then as c is an arbitrary constant, so is A
Then Archie's solution becomes $y = e^{3x}(x+A)$, which is exactly the same form as the other student's solution.

11 a)
Integrating Factor

$$= e^{\int -10 \mathrm{d}x}$$
$$= e^{-10x}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 10y = e^{2x}$$
$$\Rightarrow e^{-10x}\frac{\mathrm{d}y}{\mathrm{d}x} - 10e^{-10x}y = e^{2x} \times e^{-10x}$$
$$\Rightarrow e^{-10x}\frac{\mathrm{d}y}{\mathrm{d}x} - 10e^{-10x}y = e^{-8x}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{-10x}y) = e^{-8x}$$
$$\Rightarrow e^{-10x}y = -\frac{1}{8}e^{-8x} + c$$
$$\Rightarrow y = -\frac{1}{8}e^{2x} + ce^{10x}$$

11 b)
Integrating Factor

$$= e^{\int 7 \mathrm{d}x}$$
$$= e^{7x}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 7y = 8$$
$$\Rightarrow e^{7x}\frac{\mathrm{d}y}{\mathrm{d}x} + 7e^{7x}y = 8e^{7x}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{7x}y) = 8e^{7x}$$
$$\Rightarrow e^{7x}y = \frac{8}{7}e^{7x} + c$$
$$\Rightarrow y = \frac{8}{7} + ce^{-7x}$$

11 c)
Integrating Factor

$$= e^{\int -8\mathrm{d}x}$$
$$= e^{-8x}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 8y = 100$$
$$\Rightarrow e^{-8x}\frac{\mathrm{d}y}{\mathrm{d}x} - 8e^{-8x}y = 100e^{-8x}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{-8x}y) = 100e^{-8x}$$
$$\Rightarrow e^{-8x}y = -\frac{25}{2}e^{-8x} + c$$
$$\Rightarrow y = -\frac{25}{2} + ce^{8x}$$

11 d)
Integrating Factor

$$= e^{\int 20\mathrm{d}x}$$
$$= e^{20x}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 20y = 6e^{-12x}$$
$$\Rightarrow e^{20x}\frac{\mathrm{d}y}{\mathrm{d}x} + 20e^{20x}y = 6e^{-12x} \times e^{20x}$$
$$\Rightarrow e^{20x}\frac{\mathrm{d}y}{\mathrm{d}x} + 20e^{20x}y = 6e^{8x}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{20x}y) = 6e^{8x}$$
$$\Rightarrow e^{20x}y = \frac{3}{4}e^{8x} + c$$
$$\Rightarrow y = \frac{3}{4}e^{-12x} + ce^{-20x}$$

12 a)

$$e^{\ln x} = x$$

12 b)

$$e^{\ln x^2} = x^2$$

12 c)

$$e^{4\ln x} = e^{\ln x^4} = x^4$$

12 d)

$$e^{-3\ln x} = e^{\ln x^{-3}} = x^{-3} = \frac{1}{x^3}$$

13 a)
Integrating Factor

$$\begin{aligned} &= e^{\int \frac{1}{x}\mathrm{d}x} \\ &= e^{\ln|x|} \\ &= |x| \end{aligned}$$

Let Integrating Factor $= x$

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} &= x \\ \Rightarrow x\frac{\mathrm{d}y}{\mathrm{d}x} + y &= x^2 \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(xy) &= x^2 \\ \Rightarrow xy &= \frac{1}{3}x^3 + c \\ \Rightarrow y &= \frac{1}{3}x^2 + \frac{c}{x} \end{aligned}$$

13 b)
Integrating Factor

$$\begin{aligned} &= e^{\int \frac{2}{x}\mathrm{d}x} \\ &= e^{2\ln|x|} \\ &= e^{\ln|x^2|} \\ &= x^2 \end{aligned}$$

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} &= 30 \\ \Rightarrow x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy &= 30x^2 \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^2y) &= 30x^2 \\ \Rightarrow x^2y &= 10x^3 + c \\ \Rightarrow y &= 10x + \frac{c}{x^2} \end{aligned}$$

13 c)
Integrating Factor

$$\begin{aligned}
&= e^{\int \frac{-3}{x}\mathrm{d}x} \\
&= e^{-3\ln|x|} \\
&= e^{\ln|x^{-3}|} \\
&= \left|\frac{1}{x^3}\right|
\end{aligned}$$

Let Integrating Factor $= \frac{1}{x^3}$

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{3y}{x} &= x^3 \\
\Rightarrow \frac{1}{x^3}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{3y}{x^4} &= 1 \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{x^3}\right) &= 1 \\
\Rightarrow \frac{y}{x^3} &= x + c \\
\Rightarrow y &= x^4 + cx^3
\end{aligned}$$

13 d)
Integrating Factor

$$\begin{aligned}
&= e^{\int \frac{-5}{x}\mathrm{d}x} \\
&= e^{-5\ln|x|} \\
&= e^{\ln|x^{-5}|} \\
&= \left|\frac{1}{x^5}\right|
\end{aligned}$$

Let Integrating Factor $= \frac{1}{x^5}$

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{5y}{x} &= \frac{40}{x^4} \\
\Rightarrow \frac{1}{x^5}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{5y}{x^6} &= \frac{40}{x^9} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{x^5}\right) &= 40x^{-9} \\
\Rightarrow \frac{y}{x^5} &= -5x^{-8} + c \\
\Rightarrow y &= -\frac{5}{x^3} + cx^5
\end{aligned}$$

14 a)
Integrating Factor

$$\begin{aligned}
&= e^{\int \frac{1}{x+1}\mathrm{d}x} \\
&= e^{\ln|x+1|} \\
&= |x+1|
\end{aligned}$$

Let Integrating Factor $= x+1$

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x+1} &= 6x \\
\Rightarrow (x+1)\frac{\mathrm{d}y}{\mathrm{d}x} + y &= 6x(x+1) \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}((x+1)y) &= 6x^2 + 6x \\
\Rightarrow (x+1)y &= 2x^3 + 3x^2 + c \\
\Rightarrow y &= \frac{2x^3 + 3x^2 + c}{x+1}
\end{aligned}$$

14 b)
Integrating Factor

$$\begin{aligned}
&= e^{\int \frac{2}{x-3}\mathrm{d}x} \\
&= e^{2\ln|x-3|} \\
&= e^{\ln|(x-3)^2|} \\
&= (x-3)^2
\end{aligned}$$

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x-3} &= 12 \\
\Rightarrow (x-3)^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2(x-3)y &= 12(x-3)^2 \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}((x-3)^2 y) &= 12(x-3)^2 \\
\Rightarrow (x-3)^2 y &= 4(x-3)^3 + c \\
\Rightarrow y &= 4(x-3) + \frac{c}{(x-3)^2}
\end{aligned}$$

14 c)
Integrating Factor

$$\begin{aligned}
&= e^{\int -\frac{4}{2x+1}\mathrm{d}x} \\
&= e^{-2\ln|2x+1|} \\
&= e^{\ln|(2x+1)^{-2}|} \\
&= (2x+1)^{-2}
\end{aligned}$$

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4y}{2x+1} &= 2 \\
\Rightarrow \frac{1}{(2x+1)^2}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4y}{(2x+1)^3} &= 2(2x+1)^{-2} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{(2x+1)^2}y\right) &= 2(2x+1)^{-2} \\
\Rightarrow \frac{1}{(2x+1)^2}y &= -(2x+1)^{-1} + c \\
\Rightarrow y &= -(2x+1) + c(2x+1)^2
\end{aligned}$$

15 a)

$$p\frac{\mathrm{d}y}{\mathrm{d}x} + qy = r$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned}
p\frac{\mathrm{d}y}{\mathrm{d}x} + qy &= r \\
\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{qy}{p} &= \frac{r}{p}
\end{aligned}$$

Integrating Factor

$$\begin{aligned}
&= e^{\int \frac{q}{p}\mathrm{d}x} \\
&= e^{\frac{q}{p}x}
\end{aligned}$$

Let Integrating Factor $= e^{\frac{q}{p}x}$

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{qy}{p} &= \frac{r}{p} \\
\Rightarrow e^{\frac{q}{p}x}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{qy}{p}e^{\frac{q}{p}x} &= \frac{r}{p}e^{\frac{q}{p}x} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{\frac{q}{p}x}y) &= \frac{r}{p}e^{\frac{q}{p}x} \\
\Rightarrow e^{\frac{q}{p}x}y &= \frac{r}{p} \times \frac{p}{q}e^{\frac{q}{p}x} + c
\end{aligned}$$

$$\Rightarrow e^{\frac{q}{p}x}y = \frac{r}{q}e^{\frac{q}{p}x} + c$$
$$\Rightarrow y = \frac{r}{q} + ce^{-\frac{q}{p}x}$$

Substitute $y = \frac{2r}{q}$ when $x = 0$:

$$\frac{2r}{q} = \frac{r}{q} + ce^0$$
$$\Rightarrow \frac{2r}{q} = \frac{r}{q} + c$$
$$\Rightarrow c = \frac{r}{q}$$
$$\therefore y = \frac{r}{q} + \frac{r}{q}e^{-\frac{q}{p}x}$$

15 b)
As $x \to \infty$, $\frac{r}{q}e^{-\frac{q}{p}x} \to 0$ so $y \to \frac{r}{q}$

16 a)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + y = 1$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} = \frac{1}{x}$$

Integrating Factor

$$= e^{\int \frac{1}{x}\mathrm{d}x}$$
$$= e^{\ln|x|}$$
$$= |x|$$

Let Integrating Factor $= x$

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} = \frac{1}{x}$$
$$\Rightarrow x\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} \times x = \frac{1}{x} \times x$$
$$\Rightarrow x\frac{\mathrm{d}y}{\mathrm{d}x} + y = 1$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(xy) = 1$$
$$\Rightarrow xy = x + c$$
$$\Rightarrow y = \frac{x + c}{x}$$

16 b)

$$(x+1)\frac{\mathrm{d}y}{\mathrm{d}x} + y = 2$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x+1} = \frac{2}{x+1}$$

Integrating Factor

$$\begin{aligned} &= e^{\int \frac{1}{x+1}\mathrm{d}x} \\ &= e^{\ln|x+1|} \\ &= |x+1| \end{aligned}$$

Let Integrating Factor $= x+1$

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x+1} &= \frac{2}{x+1} \\ \Rightarrow (x+1)\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x+1} \times (x+1) &= \frac{2}{x+1} \times (x+1) \\ \Rightarrow (x+1)\frac{\mathrm{d}y}{\mathrm{d}x} + y &= 2 \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}((x+1)y) &= 2 \\ \Rightarrow (x+1)y &= 2x + c \\ \Rightarrow y &= \frac{2x+c}{x+1} \end{aligned}$$

16 c)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} - y = 1$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = \frac{1}{x}$$

Integrating Factor

$$\begin{aligned}
&= e^{\int \frac{-1}{x}\mathrm{d}x} \\
&= e^{-\ln|x|} \\
&= e^{\ln|x^{-1}|} \\
&= |x^{-1}|
\end{aligned}$$

Let Integrating Factor $= x^{-1}$

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} &= \frac{1}{x} \\
\Rightarrow x^{-1}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} \times x^{-1} &= \frac{1}{x} \times x^{-1} \\
\Rightarrow x^{-1}\frac{\mathrm{d}y}{\mathrm{d}x} - x^{-2}y &= x^{-2} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^{-1}y) &= x^{-2} \\
\Rightarrow x^{-1}y &= -x^{-1} + c \\
\Rightarrow y &= -1 + cx
\end{aligned}$$

16 d)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} - y = 2x^3$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = 2x^2$$

Integrating Factor

$$\begin{aligned} &= e^{\int \frac{-1}{x}\mathrm{d}x} \\ &= e^{-\ln|x|} \\ &= e^{\ln|x^{-1}|} \\ &= |x^{-1}| \end{aligned}$$

Let Integrating Factor $= x^{-1}$

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} &= 2x^2 \\ \Rightarrow x^{-1}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} \times x^{-1} &= 2x^2 \times x^{-1} \\ \Rightarrow x^{-1}\frac{\mathrm{d}y}{\mathrm{d}x} - x^{-2}y &= 2x \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^{-1}y) &= 2x \\ \Rightarrow x^{-1}y &= x^2 + c \\ \Rightarrow y &= x^3 + cx \end{aligned}$$

17 a)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + 3y = \sqrt{x}$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{3y}{x} &= \frac{\sqrt{x}}{x} \\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{3y}{x} &= x^{-\frac{1}{2}} \end{aligned}$$

Integrating Factor

$$\begin{aligned} &= e^{\int \frac{3}{x}\mathrm{d}x} \\ &= e^{3\ln|x|} \\ &= e^{\ln|x^3|} \\ &= |x^3| \end{aligned}$$

Let Integrating Factor $= x^3$

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{3y}{x} &= x^{-\frac{1}{2}} \\ \Rightarrow x^3\frac{\mathrm{d}y}{\mathrm{d}x} + 3x^2y &= x^{\frac{5}{2}} \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^3y) &= x^{\frac{5}{2}} \\ \Rightarrow x^3y &= \frac{2}{7}x^{\frac{7}{2}} + c \\ \Rightarrow y &= \frac{2}{7}x^{\frac{1}{2}} + cx^{-3} \end{aligned}$$

17 b)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} - y = \frac{1}{\sqrt{x}}$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} &= \frac{1}{x\sqrt{x}} \\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} &= x^{-\frac{3}{2}} \end{aligned}$$

Integrating Factor

$$\begin{aligned} &= e^{\int \frac{-1}{x}\mathrm{d}x} \\ &= e^{-\ln|x|} \\ &= e^{\ln|x^{-1}|} \\ &= \left|\frac{1}{x}\right| \end{aligned}$$

Let Integrating Factor $= \frac{1}{x}$

$$\begin{aligned} \frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x^2} &= x^{-\frac{5}{2}} \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{x}y\right) &= x^{-\frac{5}{2}} \\ \Rightarrow \frac{1}{x}y &= -\frac{2}{3}x^{-\frac{3}{2}} + c \\ \Rightarrow y &= -\frac{2}{3}x^{-\frac{1}{2}} + cx \end{aligned}$$

18 a)

$$2x^{\frac{5}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} + x^{\frac{3}{2}}y = 2$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{x^{\frac{3}{2}}y}{2x^{\frac{5}{2}}} &= \frac{2}{2x^{\frac{5}{2}}} \\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{2x}y &= x^{-\frac{5}{2}}\end{aligned}$$

Integrating Factor

$$\begin{aligned}&= e^{\int \frac{1}{2x}\mathrm{d}x} \\ &= e^{\frac{1}{2}\ln|x|} \\ &= e^{\ln|x^{\frac{1}{2}}|} \\ &= |x^{\frac{1}{2}}| \\ &= x^{\frac{1}{2}}\end{aligned}$$

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{2x}y &= x^{-\frac{5}{2}} \\ \Rightarrow x^{\frac{1}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{2x} \times x^{\frac{1}{2}} \times y &= x^{-\frac{5}{2}} \times x^{\frac{1}{2}} \\ \Rightarrow x^{\frac{1}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{2}x^{-\frac{1}{2}}y &= x^{-2} \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^{\frac{1}{2}}y) &= x^{-2} \\ \Rightarrow x^{\frac{1}{2}}y &= -x^{-1} + c \\ \Rightarrow y &= -x^{-\frac{3}{2}} + cx^{-\frac{1}{2}}\end{aligned}$$

18 b)

$$2x^{\frac{5}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} - x^{\frac{3}{2}}y = 2$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{x^{\frac{3}{2}}y}{2x^{\frac{5}{2}}} &= \frac{2}{2x^{\frac{5}{2}}} \\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{1}{2x}y &= x^{-\frac{5}{2}}\end{aligned}$$

Integrating Factor

$$\begin{aligned}&= e^{\int \frac{-1}{2x}\mathrm{d}x} \\ &= e^{-\frac{1}{2}\ln|x|} \\ &= e^{\ln|x^{-\frac{1}{2}}|} \\ &= |x^{-\frac{1}{2}}| \\ &= x^{-\frac{1}{2}}\end{aligned}$$

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{1}{2x}y &= x^{-\frac{5}{2}} \\ \Rightarrow x^{-\frac{1}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{1}{2x} \times x^{-\frac{1}{2}} \times y &= x^{-\frac{5}{2}} \times x^{-\frac{1}{2}} \\ \Rightarrow x^{-\frac{1}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{1}{2}x^{-\frac{3}{2}}y &= x^{-3} \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^{-\frac{1}{2}}y) &= x^{-3} \\ \Rightarrow x^{-\frac{1}{2}}y &= -\frac{1}{2}x^{-2} + c \\ \Rightarrow y &= -\frac{1}{2}x^{-\frac{3}{2}} + cx^{\frac{1}{2}}\end{aligned}$$

19)

$$x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 7xy = \frac{2}{x^5}$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned}x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 7xy &= \frac{2}{x^5} \\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{7y}{x} &= \frac{2}{x^7}\end{aligned}$$

Integrating Factor

$$
\begin{aligned}
&= e^{\int \frac{7}{x} \mathrm{d}x} \\
&= e^{7 \ln |x|} \\
&= e^{\ln |x^7|} \\
&= |x^7|
\end{aligned}
$$

Let Integrating Factor $= x^7$

$$
\begin{aligned}
x^7 \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{7y}{x} \times x^7 &= \frac{2}{x^7} \times x^7 \\
\Rightarrow x^7 \frac{\mathrm{d}y}{\mathrm{d}x} + 7x^6 y &= 2 \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^7 y) &= 2 \\
\Rightarrow x^7 y &= 2x + c \\
\Rightarrow y &= \frac{2}{x^6} + \frac{c}{x^7}
\end{aligned}
$$

Substitute $y = 1$ and $x = 2$:

$$
\begin{aligned}
1 &= \frac{2}{2^6} + \frac{c}{2^7} \\
\Rightarrow 1 &= \frac{1}{32} + \frac{c}{128} \\
\Rightarrow c &= 124 \\
\therefore y &= \frac{2}{x^6} + \frac{124}{x^7} \\
y &= 2x^{-6} + 124x^{-7} \\
\frac{\mathrm{d}y}{\mathrm{d}x} &= -12x^{-7} - 868x^{-8}
\end{aligned}
$$

As $x > 0$, $-12x^{-7} < 0$ and $-868x^{-8} < 0$

So $\frac{\mathrm{d}y}{\mathrm{d}x} < 0$ for all $x > 0$, and so y is a decreasing function.

20 a)

$$\sin x \frac{\mathrm{d}y}{\mathrm{d}x} - y \cos x = 2^x \sin^2 x$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned} \sin x \frac{\mathrm{d}y}{\mathrm{d}x} - y \cos x &= 2^x \sin^2 x \\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y \cos x}{\sin x} &= 2^x \sin x \\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - y \cot x &= 2^x \sin x \end{aligned}$$

Integrating Factor

$$\begin{aligned} &= e^{-\int \cot x \mathrm{d}x} \\ &= e^{-\ln|\sin x|} \\ &= e^{\ln|(\sin x)^{-1}|} \\ &= |\operatorname{cosec} x| \end{aligned}$$

Let Integrating Factor $= \operatorname{cosec} x$

$$\begin{aligned} \operatorname{cosec} x \frac{\mathrm{d}y}{\mathrm{d}x} - y \cot x \operatorname{cosec} x &= 2^x \sin x \operatorname{cosec} x \\ \Rightarrow \operatorname{cosec} x \frac{\mathrm{d}y}{\mathrm{d}x} - y \cot x \operatorname{cosec} x &= 2^x \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y \operatorname{cosec} x) &= 2^x \\ \Rightarrow y \operatorname{cosec} x &= \frac{1}{\ln 2} 2^x + c \\ \Rightarrow y &= \frac{2^x}{\ln 2} \sin x + c \sin x \end{aligned}$$

20 b)

$$\begin{aligned} 0 &= \frac{2^{\frac{\pi}{2}}}{\ln 2} \sin \frac{\pi}{2} + c \sin \frac{\pi}{2} \\ \Rightarrow 0 &= \frac{2^{\frac{\pi}{2}}}{\ln 2} + c \\ \Rightarrow c &= -\frac{2^{\frac{\pi}{2}}}{\ln 2} \\ \therefore y &= \frac{2^x}{\ln 2} \sin x - \frac{2^{\frac{\pi}{2}}}{\ln 2} \sin x \\ \Rightarrow y &= \frac{\sin x}{\ln 2} \left(2^x - 2^{\frac{\pi}{2}}\right) \end{aligned}$$

21 a)
Integrating Factor

$$= e^{\int x \mathrm{d}x}$$
$$= e^{\frac{1}{2}x^2}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} + xy = x$$
$$\Rightarrow e^{\frac{1}{2}x^2}\frac{\mathrm{d}y}{\mathrm{d}x} + xe^{\frac{1}{2}x^2}y = xe^{\frac{1}{2}x^2}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{\frac{1}{2}x^2}y) = xe^{\frac{1}{2}x^2}$$
$$\Rightarrow e^{\frac{1}{2}x^2}y = e^{\frac{1}{2}x^2} + c$$
$$\Rightarrow y = 1 + ce^{-\frac{1}{2}x^2}$$

21 b)
Integrating Factor

$$= e^{\int -6x^3 \mathrm{d}x}$$
$$= e^{-\frac{3}{2}x^4}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 6x^3y = 8x^3$$
$$\Rightarrow e^{-\frac{3}{2}x^4}\frac{\mathrm{d}y}{\mathrm{d}x} - 6x^3e^{-\frac{3}{2}x^4}y = 8x^3e^{-\frac{3}{2}x^4}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{-\frac{3}{2}x^4}y) = 8x^3e^{-\frac{3}{2}x^4}$$
$$\Rightarrow e^{-\frac{3}{2}x^4}y = -\frac{4}{3}e^{-\frac{3}{2}x^4} + c$$
$$\Rightarrow y = -\frac{4}{3} + ce^{\frac{3}{2}x^4}$$

22 a)
Integrating Factor

$$\begin{aligned} &= e^{\int \cos x \mathrm{d}x} \\ &= e^{\sin x} \end{aligned}$$

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} + y\cos x &= \cos x \\ \Rightarrow e^{\sin x}\frac{\mathrm{d}y}{\mathrm{d}x} + y\cos x e^{\sin x} &= \cos x e^{\sin x} \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{\sin x}y) &= \cos x e^{\sin x} \\ \Rightarrow e^{\sin x}y &= e^{\sin x} + c \\ \Rightarrow y &= 1 + ce^{-\sin x} \end{aligned}$$

22 b)
Integrating Factor

$$\begin{aligned} &= e^{\int -\sin 2x \mathrm{d}x} \\ &= e^{\frac{1}{2}\cos 2x} \end{aligned}$$

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} - y\sin 2x &= 4\sin 2x \\ \Rightarrow e^{\frac{1}{2}\cos 2x}\frac{\mathrm{d}y}{\mathrm{d}x} - y\sin 2x e^{\frac{1}{2}\cos 2x} &= 4\sin 2x e^{\frac{1}{2}\cos 2x} \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{\frac{1}{2}\cos 2x}y) &= 4\sin 2x e^{\frac{1}{2}\cos 2x} \\ \Rightarrow e^{\frac{1}{2}\cos 2x}y &= -4e^{\frac{1}{2}\cos 2x} + c \\ \Rightarrow y &= -4 + ce^{-\frac{1}{2}\cos 2x} \end{aligned}$$

23 a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} = 6\sqrt{x^3+2}$$

Integrating Factor

$$\begin{aligned} &= e^{\int \frac{2}{x}\mathrm{d}x} \\ &= e^{2\ln|x|} \\ &= e^{\ln|x^2|} \\ &= |x^2| \end{aligned}$$

Let Integrating Factor $= x^2$

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} &= 6\sqrt{x^3+2} \\ \Rightarrow x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy &= 6x^2\sqrt{x^3+2} \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^2y) &= 6x^2(x^3+2)^{\frac{1}{2}} \\ \Rightarrow x^2y &= \frac{4}{3}(x^3+2)^{\frac{3}{2}} + c \\ \Rightarrow y &= \frac{4(x^3+2)^{\frac{3}{2}}}{3x^2} + \frac{c}{x^2} \end{aligned}$$

23 b)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{3y}{x} = \frac{6}{\sqrt{x^4+2}}$$

Integrating Factor

$$\begin{aligned} &= e^{\int \frac{3}{x}\mathrm{d}x} \\ &= e^{3\ln|x|} \\ &= e^{\ln|x^3|} \\ &= |x^3| \end{aligned}$$

Let Integrating Factor $= x^3$

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{3y}{x} &= \frac{6}{\sqrt{x^4+2}} \\ \Rightarrow x^3\frac{\mathrm{d}y}{\mathrm{d}x} + 3x^2y &= \frac{6x^3}{\sqrt{x^4+2}} \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^3y) &= \frac{6x^3}{\sqrt{x^4+2}} \\ \Rightarrow x^3y &= 3\sqrt{x^4+2} + c \\ \Rightarrow y &= \frac{3\sqrt{x^4+2}}{x^3} + \frac{c}{x^3} \end{aligned}$$

24)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{6y}{3x-2} = 54(3x-2)^6$$

Integrating Factor

$$\begin{aligned}
&= e^{\int \frac{6}{3x-2}\mathrm{d}x} \\
&= e^{2\ln|3x-2|} \\
&= e^{\ln|(3x-2)^2|} \\
&= |(3x-2)^2|
\end{aligned}$$

Let Integrating Factor $= (3x-2)^2$

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{6y}{3x-2} &= 54(3x-2)^6 \\
\Rightarrow (3x-2)^2\frac{\mathrm{d}y}{\mathrm{d}x} + 6(3x-2)y &= 54(3x-2)^8 \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}((3x-2)^2y) &= 54(3x-2)^8 \\
\Rightarrow (3x-2)^2y &= 2(3x-2)^9 + c \\
\Rightarrow y &= 2(3x-2)^7 + c(3x-2)^{-2} \\
\frac{\mathrm{d}y}{\mathrm{d}x} &= 42(3x-2)^6 - 6c(3x-2)^{-3}
\end{aligned}$$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$
Substitute $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ when $x = \frac{1}{3}$:

$$\begin{aligned}
0 &= 42\left(3\times\frac{1}{3}-2\right)^6 - 6c\left(3\times\frac{1}{3}-2\right)^{-3} \\
\Rightarrow 0 &= 42 + 6c \\
\Rightarrow c &= -7 \\
\therefore y &= 2(3x-2)^7 - 7(3x-2)^{-2}
\end{aligned}$$

25 a)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} = 36x^6\ln x$$

Integrating Factor

$$\begin{aligned}
&= e^{\int \frac{-1}{x}\mathrm{d}x} \\
&= e^{-\ln|x|} \\
&= e^{\ln|x^{-1}|} \\
&= |x^{-1}|
\end{aligned}$$

Let Integrating Factor $= x^{-1} = \frac{1}{x}$

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x} &= 36x^6 \ln x \\ \Rightarrow \frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{x^2} &= 36x^5 \ln x \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{x}\right) &= 36x^5 \ln x \end{aligned}$$

Use Integration by Parts:

$$\begin{aligned} u &= \ln x & \frac{\mathrm{d}v}{\mathrm{d}x} &= 36x^5 \\ \frac{\mathrm{d}u}{\mathrm{d}x} &= \frac{1}{x} & v &= 6x^6 \end{aligned}$$

$$\begin{aligned} \therefore \int 36x^5 \ln x \mathrm{d}x &= 6x^6 \ln x - \int 6x^5 \mathrm{d}x \\ &= 6x^6 \ln x - x^6 + c \\ \therefore \frac{y}{x} &= 6x^6 \ln x - x^6 + c \\ \Rightarrow y &= 6x^7 \ln x - x^7 + cx \end{aligned}$$

Substitute $y = 2$ when $x = 1$:

$$\begin{aligned} 2 &= 6 \times 1^7 \times \ln 1 - 1^7 + c \times 1 \\ \Rightarrow 2 &= 0 - 1 + c \\ \Rightarrow c &= 3 \\ \therefore y &= 6x^7 \ln x - x^7 + 3x \end{aligned}$$

25 b)

$$x\sqrt{x}\frac{\mathrm{d}y}{\mathrm{d}x} + y\sqrt{x} = 8x\sqrt{x^3 + 4x}$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned} x\sqrt{x}\frac{\mathrm{d}y}{\mathrm{d}x} + y\sqrt{x} &= 8x\sqrt{x^3 + 4x} \\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{\sqrt{x}}{x\sqrt{x}}y &= \frac{8x\sqrt{x^3 + 4x}}{x\sqrt{x}} \\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{x}y &= 8\sqrt{x^2 + 4} \end{aligned}$$

Integrating Factor

$$= e^{\int \frac{1}{x}\mathrm{d}x}$$
$$= e^{\ln|x|}$$
$$= |x|$$

Let Integrating Factor $= x$

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{x}y = 8\sqrt{x^2+4}$$
$$\Rightarrow x\frac{\mathrm{d}y}{\mathrm{d}x} + y = 8x\sqrt{x^2+4}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(xy) = 8x\sqrt{x^2+4}$$
$$\Rightarrow xy = \frac{8}{3}(x^2+4)^{\frac{3}{2}} + c$$
$$\Rightarrow y = \frac{8}{3x}(x^2+4)^{\frac{3}{2}} + \frac{c}{x}$$

Substitute $y = 1$ when $x = \sqrt{5}$:

$$1 = \frac{8}{3\sqrt{5}}((\sqrt{5})^2+4)^{\frac{3}{2}} + \frac{c}{\sqrt{5}}$$
$$\Rightarrow 1 = \frac{8}{3\sqrt{5}} \times 27 + \frac{c}{\sqrt{5}}$$
$$\Rightarrow 1 = \frac{72}{\sqrt{5}} + \frac{c}{\sqrt{5}}$$
$$\Rightarrow \sqrt{5} = 72 + c$$
$$\Rightarrow c = \sqrt{5} - 72$$
$$\therefore y = \frac{8}{3x}(x^2+4)^{\frac{3}{2}} + \frac{\sqrt{5}-72}{x}$$

25 c)

$$\tan x\frac{\mathrm{d}y}{\mathrm{d}x} + y\sec^2 x = \tan^2 x$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\tan x\frac{\mathrm{d}y}{\mathrm{d}x} + y\sec^2 x = \tan^2 x$$
$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y\sec^2 x}{\tan x} = \tan x$$

Integrating Factor

$$= e^{\int \frac{\sec^2 x}{\tan x}\mathrm{d}x}$$

$$= e^{\ln|\tan x|}$$
$$= |\tan x|$$

Let Integrating Factor $= \tan x$

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y\sec^2 x}{\tan x} = \tan x$$
$$\Rightarrow \tan x\frac{\mathrm{d}y}{\mathrm{d}x} + y\sec^2 x = \tan^2 x$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\tan x) = \tan^2 x$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\tan x) = \sec^2 x - 1$$
$$\Rightarrow y\tan x = \tan x - x + c$$
$$\Rightarrow y = 1 - \frac{x}{\tan x} + \frac{c}{\tan x}$$

Substitute $y = 1$ when $x = \frac{\pi}{4}$:

$$1 = 1 - \frac{\frac{\pi}{4}}{\tan\frac{\pi}{4}} + \frac{c}{\tan\frac{\pi}{4}}$$
$$\Rightarrow 1 = 1 - \frac{\frac{\pi}{4}}{1} + \frac{c}{1}$$
$$\Rightarrow 1 = 1 - \frac{\pi}{4} + c$$
$$\Rightarrow c = \frac{\pi}{4}$$
$$\therefore y = 1 - \frac{x}{\tan x} + \frac{\frac{\pi}{4}}{\tan x}$$
$$\Rightarrow y = 1 + \frac{\frac{\pi}{4} - x}{\tan x}$$

26)

$$\frac{\mathrm{d}y}{\mathrm{d}x} = k(y - x)$$

The gradient is -8 at $(1, 3)$:

$$-8 = k(3 - 1) \Rightarrow k = -4$$
$$\therefore \frac{\mathrm{d}y}{\mathrm{d}x} = -4(y - x)$$

Rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -4y + 4x$$
$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 4x$$

Integrating Factor

$$\begin{aligned} &= e^{\int 4\mathrm{d}x} \\ &= e^{4x} \end{aligned}$$

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} + 4y &= 4x \\ \Rightarrow e^{4x}\frac{\mathrm{d}y}{\mathrm{d}x} + 4e^{4x}y &= 4xe^{4x} \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(ye^{4x}) &= 4xe^{4x} \end{aligned}$$

Use Integration by Parts:

$$\begin{aligned} u &= 4x & \frac{\mathrm{d}v}{\mathrm{d}x} &= e^{4x} \\ \frac{\mathrm{d}u}{\mathrm{d}x} &= 4 & v &= \frac{1}{4}e^{4x} \end{aligned}$$

$$\begin{aligned} \therefore \int 4xe^{4x}\mathrm{d}x &= xe^{4x} - \int e^{4x}\mathrm{d}x \\ &= xe^{4x} - \frac{1}{4}e^{4x} + c \\ \Rightarrow ye^{4x} &= xe^{4x} - \frac{1}{4}e^{4x} + c \\ \Rightarrow y &= x - \frac{1}{4} + ce^{-4x} \end{aligned}$$

Substitute $y = 3$ and $x = 1$:

$$\begin{aligned} 3 &= 1 - \frac{1}{4} + ce^{-4} \\ \Rightarrow c &= \frac{9}{4}e^{4} \\ \therefore y &= x - \frac{1}{4} + \frac{9}{4}e^{4}e^{-4x} \\ \Rightarrow y &= x - \frac{1}{4} + \frac{9}{4}e^{4-4x} \end{aligned}$$

27 a)

$$(1-\cos 2x)\frac{\mathrm{d}y}{\mathrm{d}x}+2y\sin 2x=2(\sin x+\cos x)$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x}+\mathrm{P}(x)y=\mathrm{Q}(x)$:

$$\begin{aligned}(1-\cos 2x)\frac{\mathrm{d}y}{\mathrm{d}x}+2y\sin 2x&=2(\sin x+\cos x)\\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}+\frac{2\sin 2x}{1-\cos 2x}y&=\frac{2(\sin x+\cos x)}{1-\cos 2x}\end{aligned}$$

Integrating Factor

$$\begin{aligned}&=e^{\int \frac{2\sin 2x}{1-\cos 2x}\mathrm{d}x}\\ &=e^{\ln|1-\cos 2x|}\\ &=|1-\cos 2x|\end{aligned}$$

Let Integrating Factor $=1-\cos 2x$

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{2\sin 2x}{1-\cos 2x}y&=\frac{2(\sin x+\cos x)}{1-\cos 2x}\\ \Rightarrow (1-\cos 2x)\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{2\sin 2x}{1-\cos 2x}y\times(1-\cos 2x)&=\frac{2(\sin x+\cos x)}{1-\cos 2x}\times(1-\cos 2x)\\ \Rightarrow (1-\cos 2x)\frac{\mathrm{d}y}{\mathrm{d}x}+2y\sin 2x&=2(\sin x+\cos x)\\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}((1-\cos 2x)y)&=2(\sin x+\cos x)\\ \Rightarrow (1-\cos 2x)y&=2(-\cos x+\sin x)+c\\ \Rightarrow y&=\frac{2\sin x-2\cos x+c}{1-\cos 2x}\end{aligned}$$

27 b)

$$\begin{aligned}
5 &= \frac{2\sin\frac{\pi}{2} - 2\cos\frac{\pi}{2} + c}{1 - \cos(2 \times \frac{\pi}{2})} \\
\Rightarrow 5 &= \frac{2\sin\frac{\pi}{2} - 2\cos\frac{\pi}{2} + c}{1 - \cos\pi} \\
\Rightarrow 5 &= \frac{2 - 0 + c}{1 - (-1)} \\
\Rightarrow 5 &= \frac{2 + c}{2} \\
\Rightarrow c &= 8 \\
\therefore y &= \frac{2\sin x - 2\cos x + 8}{1 - \cos 2x} \\
\Rightarrow y &= \frac{2\sin x - 2\cos x + 8}{1 - (1 - 2\sin^2 x)} \\
\Rightarrow y &= \frac{2\sin x - 2\cos x + 8}{2\sin^2 x} \\
\Rightarrow y &= \frac{\sin x - \cos x + 4}{\sin^2 x}
\end{aligned}$$

28 a)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + (2x+1)y = xe^x$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2x+1}{x}y &= e^x \\
\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \left(2 + \frac{1}{x}\right)y &= e^x
\end{aligned}$$

Integrating Factor

$$\begin{aligned}
&= e^{\int(2+\frac{1}{x})\mathrm{d}x} \\
&= e^{2x+\ln|x|} \\
&= e^{2x}e^{\ln|x|} \\
&= |x|e^{2x}
\end{aligned}$$

Let Integrating Factor $= xe^{2x}$

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} + \left(2 + \frac{1}{x}\right)y &= e^x \\
\Rightarrow xe^{2x}\frac{\mathrm{d}y}{\mathrm{d}x} + \left(2 + \frac{1}{x}\right)xe^{2x}y &= e^x \times xe^{2x} \\
\Rightarrow xe^{2x}\frac{\mathrm{d}y}{\mathrm{d}x} + (2x+1)e^{2x}y &= xe^{3x}
\end{aligned}$$

$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(xe^{2x}y) = xe^{3x}$$

Use Integration by Parts:

$$\begin{aligned} u &= x & \frac{\mathrm{d}v}{\mathrm{d}x} &= e^{3x} \\ \frac{\mathrm{d}u}{\mathrm{d}x} &= 1 & v &= \frac{1}{3}e^{3x} \end{aligned}$$

$$\begin{aligned} \therefore \int xe^{3x}\mathrm{d}x &= \frac{1}{3}xe^{3x} - \int \frac{1}{3}e^{3x}\mathrm{d}x \\ \Rightarrow \int xe^{3x}\mathrm{d}x &= \frac{1}{3}xe^{3x} - \frac{1}{9}e^{3x} + c \end{aligned}$$

So:

$$\begin{aligned} xe^{2x}y &= \frac{1}{3}xe^{3x} - \frac{1}{9}e^{3x} + c \\ \Rightarrow y &= \frac{xe^{3x}}{3xe^{2x}} - \frac{e^{3x}}{9xe^{2x}} + \frac{c}{xe^{2x}} \\ \Rightarrow y &= \frac{1}{3}e^{x} - \frac{1}{9x}e^{x} + \frac{c}{xe^{2x}} \\ \Rightarrow y &= \left(\frac{1}{3} - \frac{1}{9x}\right)e^{x} + \frac{c}{xe^{2x}} \\ \Rightarrow y &= \frac{3x-1}{9x}e^{x} + \frac{c}{xe^{2x}} \end{aligned}$$

28 b)

$$\begin{aligned} \frac{1}{3}e &= \frac{3\times 1 - 1}{9\times 1}e^{1} + \frac{c}{1\times e^{2\times 1}} \\ \Rightarrow \frac{1}{3}e &= \frac{2}{9}e + \frac{c}{e^{2}} \\ \Rightarrow \frac{1}{9}e &= \frac{c}{e^{2}} \\ \Rightarrow c &= \frac{1}{9}e^{3} \\ \therefore y &= \frac{3x-1}{9x}e^{x} + \frac{\frac{1}{9}e^{3}}{xe^{2x}} \\ \Rightarrow y &= \frac{3x-1}{9x}e^{x} + \frac{e^{3-2x}}{9x} \\ \Rightarrow y &= \frac{(3x-1)e^{x} + e^{3-2x}}{9x} \end{aligned}$$

29)

$$\frac{\mathrm{d}y}{\mathrm{d}x} - y\tan x = \sin 2x$$

Integrating Factor

$$\begin{aligned}
&= e^{\int -\tan x \mathrm{d}x} \\
&= e^{-\ln|\sec x|} \\
&= e^{\ln|(\sec x)^{-1}|} \\
&= e^{\ln|\cos x|} \\
&= |\cos x|
\end{aligned}$$

Let Integrating Factor $= \cos x$

$$\begin{aligned}
\cos x\frac{\mathrm{d}y}{\mathrm{d}x} - y\tan x \times \cos x &= \sin 2x \times \cos x \\
\Rightarrow \cos x\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y\sin x}{\cos x} \times \cos x &= 2\sin x\cos x \times \cos x \\
\Rightarrow \cos x\frac{\mathrm{d}y}{\mathrm{d}x} - y\sin x &= 2\sin x\cos^2 x \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\cos x) &= 2\sin x\cos^2 x \\
\Rightarrow y\cos x &= -\frac{2}{3}\cos^3 x + c \\
\Rightarrow y &= -\frac{2}{3}\cos^2 x + c\sec x
\end{aligned}$$

Substitute $y = \frac{4}{3}$ when $x = 0$:

$$\begin{aligned}
\frac{4}{3} &= -\frac{2}{3}\cos^2 0 + c\sec 0 \\
\Rightarrow \frac{4}{3} &= -\frac{2}{3} + c \\
\Rightarrow c &= 2 \\
\therefore y &= -\frac{2}{3}\cos^2 x + 2\sec x
\end{aligned}$$

30 a)

$$\sec x \frac{\mathrm{d}y}{\mathrm{d}x} + y \operatorname{cosec} x = 2 \tan x$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned} \frac{1}{\cos x}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{\sin x} &= \frac{2\sin x}{\cos x} \\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y\cos x}{\sin x} &= 2\sin x \end{aligned}$$

Integrating Factor

$$\begin{aligned} &= e^{\int \frac{\cos x}{\sin x}\mathrm{d}x} \\ &= e^{\ln|\sin x|} \\ &= |\sin x| \end{aligned}$$

Let Integrating Factor $= \sin x$

$$\begin{aligned} \sin x \frac{\mathrm{d}y}{\mathrm{d}x} + y\cos x &= 2\sin^2 x \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\sin x) &= 2\sin^2 x \\ \cos 2A \equiv 1 - 2\sin^2 A \Rightarrow \sin^2 A &\equiv \frac{1}{2} - \frac{1}{2}\cos 2A \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\sin x) &= 2\left(\frac{1}{2} - \frac{1}{2}\cos 2x\right) \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\sin x) &= 1 - \cos 2x \\ \Rightarrow y\sin x &= x - \frac{1}{2}\sin 2x + c \\ \Rightarrow y &= \frac{x}{\sin x} - \frac{\sin 2x}{2\sin x} + \frac{c}{\sin x} \\ \Rightarrow y &= x\operatorname{cosec} x - \frac{2\sin x\cos x}{2\sin x} + c\operatorname{cosec} x \\ \Rightarrow y &= x\operatorname{cosec} x - \cos x + c\operatorname{cosec} x \end{aligned}$$

30 b)

$$\begin{aligned}
\frac{12}{\sqrt{3}} &= \frac{\pi}{3} \times \operatorname{cosec}\frac{\pi}{3} - \cos\frac{\pi}{3} + c\operatorname{cosec}\frac{\pi}{3} \\
\Rightarrow 4\sqrt{3} &= \frac{2\pi\sqrt{3}}{9} - \frac{1}{2} + \frac{2c\sqrt{3}}{3} \\
\Rightarrow \frac{2c\sqrt{3}}{3} &= 4\sqrt{3} - \frac{2\pi\sqrt{3}}{9} + \frac{1}{2} \\
\Rightarrow c &= 6 - \frac{\pi}{3} + \frac{\sqrt{3}}{4} \\
\therefore y &= x\operatorname{cosec} x - \cos x + \left(6 - \frac{\pi}{3} + \frac{\sqrt{3}}{4}\right)\operatorname{cosec} x
\end{aligned}$$

31)

$$x^4\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{x^3y}{\ln x} = 3$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned}
x^4\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{x^3y}{\ln x} &= 3 \\
\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x\ln x} &= \frac{3}{x^4}
\end{aligned}$$

Integrating Factor

$$\begin{aligned}
&= e^{\int \frac{1}{x\ln x}\mathrm{d}x} \\
&= e^{\ln|\ln x|} \\
&= |\ln x|
\end{aligned}$$

Let Integrating Factor $= \ln x$

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x\ln x} &= \frac{3}{x^4} \\
\Rightarrow \ln x\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x\ln x} \times \ln x &= \frac{3}{x^4} \times \ln x \\
\Rightarrow \ln x\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} &= \frac{3\ln x}{x^4} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\ln x) &= 3x^{-4}\ln x
\end{aligned}$$

Use Integration by Parts:

$$u = \ln x \qquad\qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 3x^{-4}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} \qquad\qquad v = -x^{-3}$$

$$\begin{aligned}
\therefore \int 3x^{-4}\ln x \mathrm{d}x &= -x^{-3}\ln x + \int x^{-4}\mathrm{d}x \\
&= -x^{-3}\ln x - \frac{1}{3}x^{-3} + c \\
\Rightarrow y\ln x &= -x^{-3}\ln x - \frac{1}{3}x^{-3} + c \\
\Rightarrow y &= -x^{-3} - \frac{\frac{1}{3}x^{-3}}{\ln x} + \frac{c}{\ln x}
\end{aligned}$$

Substitute $y = e^{-3}$ when $x = e$:

$$\begin{aligned}
e^{-3} &= -e^{-3} - \frac{\frac{1}{3}e^{-3}}{\ln e} + \frac{c}{\ln e} \\
\Rightarrow e^{-3} &= -e^{-3} - \frac{\frac{1}{3}e^{-3}}{1} + \frac{c}{1} \\
\Rightarrow e^{-3} &= -e^{-3} - \frac{1}{3}e^{-3} + c \\
\Rightarrow c &= \frac{7}{3}e^{-3} \\
\therefore y &= -x^{-3} - \frac{\frac{1}{3}x^{-3}}{\ln x} + \frac{\frac{7}{3}e^{-3}}{\ln x} \\
\Rightarrow y &= -\frac{1}{x^3} - \frac{1}{3x^3\ln x} + \frac{7e^{-3}}{3\ln x} \\
\Rightarrow y &= -\frac{3\ln x}{3x^3\ln x} - \frac{1}{3x^3\ln x} + \frac{7e^{-3}x^3}{3x^3\ln x} \\
\Rightarrow y &= \frac{7e^{-3}x^3 - 3\ln x - 1}{3x^3\ln x}
\end{aligned}$$

32 a)
Integrating Factor

$$\begin{aligned}
&= e^{\int \sinh x \mathrm{d}x} \\
&= e^{\cosh x}
\end{aligned}$$

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} + y\sinh x &= 2\sinh x \\
\Rightarrow e^{\cosh x}\frac{\mathrm{d}y}{\mathrm{d}x} + y\sinh x e^{\cosh x} &= 2\sinh x e^{\cosh x} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{\cosh x}y) &= 2\sinh x e^{\cosh x} \\
\Rightarrow e^{\cosh x}y &= 2e^{\cosh x} + c \\
\Rightarrow y &= 2 + ce^{-\cosh x}
\end{aligned}$$

32 b)
Integrating Factor

$$= e^{\int -2\cosh 3x \mathrm{d}x}$$
$$= e^{-\frac{2}{3}\sinh 3x}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 2y\cosh 3x = 6\cosh 3x$$
$$\Rightarrow e^{-\frac{2}{3}\sinh 3x}\frac{\mathrm{d}y}{\mathrm{d}x} - 2y\cosh 3xe^{-\frac{2}{3}\sinh 3x} = 6\cosh 3xe^{-\frac{2}{3}\sinh 3x}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{-\frac{2}{3}\sinh 3x}y) = 6\cosh 3xe^{-\frac{2}{3}\sinh 3x}$$
$$\Rightarrow e^{-\frac{2}{3}\sinh 3x}y = -3e^{-\frac{2}{3}\sinh 3x} + c$$
$$\Rightarrow y = -3 + ce^{\frac{2}{3}\sinh 3x}$$

32 c)

$$2\cosh^2 x\frac{\mathrm{d}y}{\mathrm{d}x} + y\sinh 2x = \sinh^5 x\cosh^2 x$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y\sinh 2x}{2\cosh^2 x} &= \frac{\sinh^5 x\cosh^2 x}{2\cosh^2 x}\\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y\sinh x\cosh x}{2\cosh^2 x} &= \frac{1}{2}\sinh^5 x\\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + y\tanh x &= \frac{1}{2}\sinh^5 x\end{aligned}$$

Integrating Factor

$$\begin{aligned}&= e^{\int \tanh x\mathrm{d}x}\\ &= e^{\ln|\cosh x|}\\ &= |\cosh x|\end{aligned}$$

Let Integrating Factor $= \cosh x$

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} + y\tanh x &= \frac{1}{2}\sinh^5 x\\ \Rightarrow \cosh x\frac{\mathrm{d}y}{\mathrm{d}x} + y\tanh x\cosh x &= \frac{1}{2}\sinh^5 x\cosh x\\ \Rightarrow \cosh x\frac{\mathrm{d}y}{\mathrm{d}x} + y\sinh x &= \frac{1}{2}\sinh^5 x\cosh x\\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\cosh x) &= \frac{1}{2}\sinh^5 x\cosh x\\ \Rightarrow y\cosh x &= \frac{1}{12}\sinh^6 x + k\\ \Rightarrow y &= \frac{\frac{1}{12}\sinh^6 x + k}{\cosh x}\\ \Rightarrow y &= \frac{\sinh^6 x + c}{12\cosh x}\end{aligned}$$

where $c = 12k$

33)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = \frac{7-x}{(x^2+7)(x+1)}$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned}x\frac{\mathrm{d}y}{\mathrm{d}x} + 2y &= \frac{7-x}{(x^2+7)(x+1)}\\ \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} &= \frac{7-x}{x(x^2+7)(x+1)}\end{aligned}$$

Integrating Factor

$$\begin{aligned}
&= e^{\int \frac{2}{x}\mathrm{d}x} \\
&= e^{2\ln|x|} \\
&= e^{\ln|x^2|} \\
&= x^2
\end{aligned}$$

$$\begin{aligned}
x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy &= \frac{7x - x^2}{(x^2+7)(x+1)} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^2y) &= \frac{7x - x^2}{(x^2+7)(x+1)}
\end{aligned}$$

Use Partial Fractions:

$$\begin{aligned}
\frac{7x - x^2}{(x^2+7)(x+1)} &\equiv \frac{Ax+B}{x^2+7} + \frac{C}{x+1} \\
\Rightarrow 7x - x^2 &\equiv (Ax+B)(x+1) + C(x^2+7)
\end{aligned}$$

Let $x = -1$: $-8 = 8C \Rightarrow C = -1$

Let $x = 0$: $0 = B - 7 \Rightarrow B = 7$

Let $x = 1$: $6 = (A+7) \times 2 - 8 \Rightarrow A = 0$

$$\therefore \frac{7x - x^2}{(x^2+7)(x+1)} \equiv \frac{7}{x^2+7} - \frac{1}{x+1}$$

So:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(x^2y) &= \frac{7}{x^2+7} - \frac{1}{x+1} \\
\frac{\mathrm{d}}{\mathrm{d}x}(x^2y) &= \frac{7}{7(1+\frac{1}{7}x^2)} - \frac{1}{x+1} \\
\frac{\mathrm{d}}{\mathrm{d}x}(x^2y) &= \frac{1}{1+(\frac{1}{\sqrt{7}}x)^2} - \frac{1}{x+1} \\
\Rightarrow x^2y &= \frac{1}{\frac{1}{\sqrt{7}}}\arctan\left(\frac{1}{\sqrt{7}}x\right) - \ln|x+1| + c \\
\Rightarrow x^2y &= \sqrt{7}\arctan\left(\frac{1}{\sqrt{7}}x\right) - \ln|x+1| + c \\
\Rightarrow y &= \frac{\sqrt{7}}{x^2}\arctan\left(\frac{x}{\sqrt{7}}\right) - \frac{\ln|x+1|}{x^2} + \frac{c}{x^2}
\end{aligned}$$

34 a)

$$
\begin{aligned}
& e^{\int \frac{1}{1-x^2}\,\mathrm{d}x} \\
&= e^{\operatorname{artanh} x} \\
&= e^{\frac{1}{2}\ln(\frac{1+x}{1-x})} \\
&= e^{\ln((\frac{1+x}{1-x})^{\frac{1}{2}})} \\
&= \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} \quad \text{as required}
\end{aligned}
$$

34 b)

$$(1-x^2)\frac{\mathrm{d}y}{\mathrm{d}x} + y = (1-x)^{\frac{3}{2}}(1+x)$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$
\begin{aligned}
& (1-x^2)\frac{\mathrm{d}y}{\mathrm{d}x} + y = (1-x)^{\frac{3}{2}}(1+x) \\
\Rightarrow\ & \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{1-x^2} = \frac{(1-x)^{\frac{3}{2}}(1+x)}{1-x^2} \\
\Rightarrow\ & \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{1-x^2} = \frac{(1-x)^{\frac{3}{2}}(1+x)}{(1-x)(1+x)} \\
\Rightarrow\ & \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{1-x^2} = (1-x)^{\frac{1}{2}}
\end{aligned}
$$

Integrating Factor

$$
\begin{aligned}
&= e^{\int \frac{1}{1-x^2}\,\mathrm{d}x} \\
&= \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}
\end{aligned}
$$

$$
\begin{aligned}
& \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{1-x^2} = (1-x)^{\frac{1}{2}} \\
\Rightarrow\ & \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{1-x^2}\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} = (1-x)^{\frac{1}{2}}\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} \\
\Rightarrow\ & \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{1-x^2}\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} = (1-x)^{\frac{1}{2}} \times \frac{(1+x)^{\frac{1}{2}}}{(1-x)^{\frac{1}{2}}} \\
\Rightarrow\ & \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{1-x^2}\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} = (1+x)^{\frac{1}{2}} \\
& \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} y\right) = (1+x)^{\frac{1}{2}} \\
& \Rightarrow \left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} y = \frac{2}{3}(1+x)^{\frac{3}{2}} + c
\end{aligned}
$$

$$\Rightarrow \frac{(1+x)^{\frac{1}{2}}}{(1-x)^{\frac{1}{2}}}y = \frac{2}{3}(1+x)^{\frac{3}{2}} + c$$

$$\Rightarrow y = \frac{2}{3}(1+x)(1-x)^{\frac{1}{2}} + \frac{c(1-x)^{\frac{1}{2}}}{(1+x)^{\frac{1}{2}}}$$

Substitute $y = \frac{5}{3}$ when $x = 0$:

$$\begin{aligned}
\frac{5}{3} &= \frac{2}{3}(1+0)(1-0)^{\frac{1}{2}} + \frac{c(1-0)^{\frac{1}{2}}}{(1+0)^{\frac{1}{2}}} \\
\Rightarrow \frac{5}{3} &= \frac{2}{3} + c \\
\Rightarrow c &= 1 \\
\therefore y &= \frac{2}{3}(1+x)(1-x)^{\frac{1}{2}} + \left(\frac{1-x}{1+x}\right)^{\frac{1}{2}}
\end{aligned}$$

35)

$$(9-4x^2)\frac{\mathrm{d}y}{\mathrm{d}x} - 4xy = 2$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$\begin{aligned}
(9-4x^2)\frac{\mathrm{d}y}{\mathrm{d}x} - 4xy &= 2 \\
\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4x}{9-4x^2}y &= \frac{2}{9-4x^2}
\end{aligned}$$

Integrating Factor

$$\begin{aligned}
&= e^{\int -\frac{4x}{9-4x^2}\mathrm{d}x} \\
&= e^{\frac{1}{2}\ln|9-4x^2|} \\
&= e^{\ln|(9-4x^2)^{\frac{1}{2}}|} \\
&= |\sqrt{9-4x^2}|
\end{aligned}$$

Let Integrating Factor $= \sqrt{9-4x^2}$

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4x}{9-4x^2}y &= \frac{2}{9-4x^2} \\
\Rightarrow \sqrt{9-4x^2}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4x}{9-4x^2}y \times \sqrt{9-4x^2} &= \frac{2}{9-4x^2} \times \sqrt{9-4x^2} \\
\Rightarrow \sqrt{9-4x^2}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{4x}{\sqrt{9-4x^2}}y &= \frac{2}{\sqrt{9-4x^2}} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\sqrt{9-4x^2}) &= \frac{2}{\sqrt{9-4x^2}}
\end{aligned}$$

$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\sqrt{9-4x^2}) = \frac{2}{\sqrt{9(1-\frac{4}{9}x^2)}}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\sqrt{9-4x^2}) = \frac{2}{3\sqrt{1-(\frac{2}{3}x)^2}}$$
$$\Rightarrow y\sqrt{9-4x^2} = \frac{2}{3} \times \frac{1}{\frac{2}{3}} \times \arcsin\left(\frac{2}{3}x\right) + c$$
$$\Rightarrow y\sqrt{9-4x^2} = \arcsin\left(\frac{2}{3}x\right) + c$$
$$\Rightarrow y = \frac{\arcsin(\frac{2}{3}x) + c}{\sqrt{9-4x^2}}$$

Substitute $y = 0$ when $x = \frac{3\sqrt{2}}{4}$:

$$0 = \frac{\arcsin(\frac{2}{3} \times \frac{3\sqrt{2}}{4}) + c}{\sqrt{9-4(\frac{3\sqrt{2}}{4})^2}}$$
$$\Rightarrow 0 = \frac{\arcsin(\frac{\sqrt{2}}{2}) + c}{\sqrt{\frac{9}{2}}}$$
$$\Rightarrow 0 = \frac{\pi}{4} + c$$
$$\Rightarrow c = -\frac{\pi}{4}$$
$$\therefore y = \frac{\arcsin(\frac{2}{3}x) - \frac{\pi}{4}}{\sqrt{9-4x^2}}$$

36)

$$2x\frac{\mathrm{d}y}{\mathrm{d}x} - y = \sqrt{\frac{4x^3}{4x-x^2-1}}$$

First, rearrange into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$:

$$2x\frac{\mathrm{d}y}{\mathrm{d}x} - y = \sqrt{\frac{4x^3}{4x-x^2-1}}$$
$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x} = \frac{1}{2x}\sqrt{\frac{4x^3}{4x-x^2-1}}$$
$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x} = \frac{1}{2x}\sqrt{\frac{4x^2 \times x}{4x-x^2-1}}$$
$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x} = \sqrt{\frac{x}{4x-x^2-1}}$$

Integrating Factor

$$= e^{\int \frac{-1}{2x}\mathrm{d}x}$$

$$= e^{-\frac{1}{2}\int \frac{1}{x}\mathrm{d}x}$$
$$= e^{-\frac{1}{2}\ln|x|}$$
$$= e^{\ln|x^{-\frac{1}{2}}|}$$
$$= |x^{-\frac{1}{2}}|$$

Let Integrating Factor $= x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}$

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x} = \sqrt{\frac{x}{4x - x^2 - 1}}$$
$$\Rightarrow \frac{1}{\sqrt{x}}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x} \times \frac{1}{\sqrt{x}} = \frac{1}{\sqrt{x}} \times \sqrt{\frac{x}{4x - x^2 - 1}}$$
$$\Rightarrow \frac{1}{\sqrt{x}}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x^{\frac{3}{2}}} = \frac{1}{\sqrt{x}} \times \frac{\sqrt{x}}{\sqrt{4x - x^2 - 1}}$$
$$\Rightarrow \frac{1}{\sqrt{x}}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{y}{2x^{\frac{3}{2}}} = \frac{1}{\sqrt{4x - x^2 - 1}}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sqrt{x}}\right) = \frac{1}{\sqrt{4x - x^2 - 1}}$$

Now, $4x - x^2 - 1 \equiv -(x^2 - 4x) - 1 \equiv -((x-2)^2 - 4) - 1 \equiv -(x-2)^2 + 4 - 1$

$$\equiv 3 - (x-2)^2$$

So:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sqrt{x}}\right) = \frac{1}{\sqrt{3 - (x-2)^2}}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sqrt{x}}\right) = \frac{1}{\sqrt{3(1 - \frac{1}{3}(x-2)^2)}}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sqrt{x}}\right) = \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{1 - (\frac{x-2}{\sqrt{3}})^2}}$$
$$\Rightarrow \frac{y}{\sqrt{x}} = \frac{1}{\sqrt{3}} \times \frac{1}{\frac{1}{\sqrt{3}}} \times \arcsin\left(\frac{x-2}{\sqrt{3}}\right) + c$$
$$\Rightarrow y = \sqrt{x}\arcsin\left(\frac{x-2}{\sqrt{3}}\right) + c\sqrt{x}$$

Substitute $y = \sqrt{\frac{7}{2}}$ when $x = \frac{7}{2}$:

$$\sqrt{\frac{7}{2}} = \sqrt{\frac{7}{2}}\arcsin\left(\frac{\frac{7}{2} - 2}{\sqrt{3}}\right) + c\sqrt{\frac{7}{2}}$$
$$\Rightarrow 1 = \arcsin\frac{\sqrt{3}}{2} + c$$
$$\Rightarrow 1 = \frac{\pi}{3} + c$$
$$\Rightarrow c = 1 - \frac{\pi}{3}$$

$$\therefore y = \sqrt{x}\arcsin\left(\frac{x-2}{\sqrt{3}}\right) + \left(1 - \frac{\pi}{3}\right)\sqrt{x}$$

37 a)
Begin with the homogeneous DE:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 0$$

Assume $y = e^{rx}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = re^{rx}$
Then $\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 0$ becomes:

$$\begin{aligned} re^{rx} + 6e^{rx} &= 0 \\ \Rightarrow r + 6 &= 0 \\ \Rightarrow r &= -6 \\ \therefore y &= ce^{-6x} \end{aligned}$$

Assume a particular integral of the form $y = Ae^{5x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 5Ae^{5x}$
Then substituting into $\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = e^{5x}$ we get:

$$5Ae^{5x} + 6Ae^{5x} = e^{5x} \Rightarrow 5A + 6A = 1 \Rightarrow 11A = 1 \Rightarrow A = \frac{1}{11}$$

$\therefore$ The General solution is

$$y = ce^{-6x} + \frac{1}{11}e^{5x}$$

37 b)
Begin with the homogeneous DE:

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = 0$$

Assume $y = e^{rx}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = re^{rx}$
Then $\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = 0$ becomes:

$$\begin{aligned} re^{rx} - 3e^{rx} &= 0 \\ \Rightarrow r - 3 &= 0 \\ \Rightarrow r &= 3 \\ \therefore y &= ce^{3x} \end{aligned}$$

Assume a particular integral of the form $y = Ax + B \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = A$
Then substituting into $\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = x$ we get:

$$A - 3(Ax + B) = x \Rightarrow -3Ax + (A - 3B) = x$$

Comparing coefficients: $-3A = 1 \Rightarrow A = -\frac{1}{3}$ and $-\frac{1}{3} - 3B = 0 \Rightarrow B = -\frac{1}{9}$
$\therefore$ The General solution is

$$y = ce^{3x} - \frac{1}{3}x - \frac{1}{9}$$

37 c)
Begin with the homogeneous DE:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$$

Assume $y = e^{rx}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = re^{rx}$
Then $\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$ becomes:

$$\begin{aligned} re^{rx} + 4e^{rx} &= 0 \\ \Rightarrow r + 4 &= 0 \\ \Rightarrow r &= -4 \\ \therefore y &= ce^{-4x} \end{aligned}$$

Assume a particular integral of the form $y = A\sin x + B\cos x$

$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = A\cos x - B\sin x$$

Then substituting into $\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 17\sin x$ we get:

$$\begin{aligned} A\cos x - B\sin x + 4(A\sin x + B\cos x) &= 17\sin x \\ \Rightarrow (4A - B)\sin x + (A + 4B)\cos x &= 17\sin x \end{aligned}$$

Comparing coefficients:

$$\begin{aligned} 4A - B &= 17 \quad (1) \\ A + 4B &= 0 \quad (2) \end{aligned}$$

So $A = 4$ and $B = -1$
$\therefore$ The General solution is

$$y = ce^{-4x} + 4\sin x - \cos x$$

37 d)
Begin with the homogeneous DE:

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 7y = 0$$

Assume $y = e^{rx}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = re^{rx}$
Then $\frac{\mathrm{d}y}{\mathrm{d}x} - 7y = 0$ becomes:

$$\begin{aligned} re^{rx} - 7e^{rx} &= 0 \\ \Rightarrow r - 7 &= 0 \\ \Rightarrow r &= 7 \\ \therefore y &= ce^{7x} \end{aligned}$$

Assume a particular integral of the form $y = Axe^{7x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 7Axe^{7x} + Ae^{7x}$
Then substituting into $\frac{\mathrm{d}y}{\mathrm{d}x} - 7y = e^{7x}$ we get:

$$7Axe^{7x} + Ae^{7x} - 7Axe^{7x} = e^{7x} \Rightarrow Ae^{7x} = e^{7x} \Rightarrow A = 1$$

$\therefore$ The General solution is

$$y = ce^{7x} + xe^{7x}$$

38 a)
Assume a particular integral of the form $y = Ax + B + C\sin x + D\cos x$

$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = A + C\cos x - D\sin x$$

Then substituting into $\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 15\cos x - 4x$ we get:

$$\begin{aligned} A + C\cos x - D\sin x + 2(Ax + B + C\sin x + D\cos x) &= 15\cos x - 4x \\ \Rightarrow 2Ax + (A + 2B) + (C + 2D)\cos x + (2C - D)\sin x &= -4x + 15\cos x \\ 2A &= -4 \Rightarrow A = -2 \\ -2 + 2B &= 0 \Rightarrow B = 1 \\ C + 2D &= 15 \quad (1) \\ 2C - D &= 0 \quad (2) \end{aligned}$$

$C = 3$ and $D = 6$
$\therefore$ A particular integral is $y = -2x + 1 + 3\sin x + 6\cos x$

38 b)
Begin with the homogeneous DE:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0$$

Assume $y = e^{rx}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = re^{rx}$
Then $\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0$ becomes:

$$\begin{aligned} re^{rx} + 2e^{rx} &= 0 \\ \Rightarrow r + 2 &= 0 \\ \Rightarrow r &= -2 \\ \therefore y &= ce^{-2x} \end{aligned}$$

So the General solution is

$$y = ce^{-2x} - 2x + 1 + 3\sin x + 6\cos x$$

39)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y = xy^2$$

Let $u = \frac{1}{y} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{y^2}\frac{\mathrm{d}y}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -y^2\frac{\mathrm{d}u}{\mathrm{d}x}$
$u = \frac{1}{y} \Rightarrow y = \frac{1}{u}$

$$\begin{aligned} \therefore -y^2\frac{\mathrm{d}u}{\mathrm{d}x} + y &= xy^2 \\ \Rightarrow -\frac{1}{u^2}\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{1}{u} &= \frac{x}{u^2} \end{aligned}$$

Rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x} + \mathrm{P}(x)u = \mathrm{Q}(x)$:

$$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} - u = -x$$

Integrating Factor

$$\begin{aligned} &= e^{\int -1\mathrm{d}x} \\ &= e^{-x} \end{aligned}$$

$$\begin{aligned} \frac{\mathrm{d}u}{\mathrm{d}x} - u &= -x \\ \Rightarrow e^{-x}\frac{\mathrm{d}u}{\mathrm{d}x} - ue^{-x} &= -xe^{-x} \end{aligned}$$

$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(ue^{-x}) = -xe^{-x}$$

Use Integration by Parts:

$$\begin{aligned} u &= -x & \frac{\mathrm{d}v}{\mathrm{d}x} &= e^{-x} \\ \frac{\mathrm{d}u}{\mathrm{d}x} &= -1 & v &= -e^{-x} \end{aligned}$$

$$\begin{aligned} \therefore \int -xe^{-x}\mathrm{d}x &= xe^{-x} - \int e^{-x}\mathrm{d}x \\ &= xe^{-x} + e^{-x} + c \\ \Rightarrow ue^{-x} &= xe^{-x} + e^{-x} + c \\ \Rightarrow u &= x + 1 + ce^{x} \\ \Rightarrow \frac{1}{y} &= x + 1 + ce^{x} \\ \Rightarrow y &= \frac{1}{x + 1 + ce^{x}} \end{aligned}$$

40)

$$5xy^4\frac{\mathrm{d}y}{\mathrm{d}x} + 3y^5 = \frac{\sin 2x}{x^2}$$

Let $u = y^5 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 5y^4\frac{\mathrm{d}y}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{5y^4}\frac{\mathrm{d}u}{\mathrm{d}x}$

$$\begin{aligned} \therefore 5xy^4 \times \frac{1}{5y^4}\frac{\mathrm{d}u}{\mathrm{d}x} + 3u &= \frac{\sin 2x}{x^2} \\ \Rightarrow x\frac{\mathrm{d}u}{\mathrm{d}x} + 3u &= \frac{\sin 2x}{x^2} \end{aligned}$$

Rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x} + \mathrm{P}(x)u = \mathrm{Q}(x)$:

$$\begin{aligned} x\frac{\mathrm{d}u}{\mathrm{d}x} + 3u &= \frac{\sin 2x}{x^2} \\ \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} + \frac{3u}{x} &= \frac{\sin 2x}{x^3} \end{aligned}$$

Integrating Factor

$$\begin{aligned} &= e^{\int \frac{3}{x}\mathrm{d}x} \\ &= e^{3\ln|x|} \\ &= e^{\ln|x^3|} \\ &= |x^3| \end{aligned}$$

Let Integrating Factor $= x^3$

$$\begin{aligned}
\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{3u}{x} &= \frac{\sin 2x}{x^3} \\
\Rightarrow x^3\frac{\mathrm{d}u}{\mathrm{d}x} + 3x^2u &= \sin 2x \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^3u) &= \sin 2x \\
\Rightarrow x^3u &= -\frac{1}{2}\cos 2x + k \\
\Rightarrow u &= \frac{-\frac{1}{2}\cos 2x + k}{x^3} \\
\Rightarrow y^5 &= \frac{k - \frac{1}{2}\cos 2x}{x^3} \\
\Rightarrow y &= \sqrt[5]{\frac{k - \frac{1}{2}\cos 2x}{x^3}} \\
\Rightarrow y &= \sqrt[5]{\frac{c - \cos 2x}{2x^3}}
\end{aligned}$$

where $c = 2k$

41)

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 4xy^3 = y$$

Let $u = \frac{1}{y^2} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{2}{y^3}\frac{\mathrm{d}y}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{2}y^3\frac{\mathrm{d}u}{\mathrm{d}x}$

$$\begin{aligned}
\therefore -\frac{1}{2}y^3\frac{\mathrm{d}u}{\mathrm{d}x} + 4xy^3 &= y \\
\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} - 8x &= -\frac{2}{y^2} \\
\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} - 8x &= -2u
\end{aligned}$$

Rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x} + \mathrm{P}(x)u = \mathrm{Q}(x)$:

$$\frac{\mathrm{d}u}{\mathrm{d}x} + 2u = 8x$$

Integrating Factor

$$\begin{aligned}
&= e^{\int 2\mathrm{d}x} \\
&= e^{2x}
\end{aligned}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} + 2u = 8x$$

$$\Rightarrow e^{2x}\frac{\mathrm{d}u}{\mathrm{d}x} + 2e^{2x}u = 8xe^{2x}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(e^{2x}u) = 8xe^{2x}$$

Use Integration by Parts:

$$u = 8x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = e^{2x}$$
$$\frac{\mathrm{d}u}{\mathrm{d}x} = 8 \qquad v = \frac{1}{2}e^{2x}$$

$$\therefore \int 8xe^{2x}\mathrm{d}x = 4xe^{2x} - \int 4e^{2x}\mathrm{d}x$$
$$= 4xe^{2x} - 2e^{2x} + c$$
$$\Rightarrow e^{2x}u = 4xe^{2x} - 2e^{2x} + c$$
$$\Rightarrow u = 4x - 2 + ce^{-2x}$$
$$\Rightarrow \frac{1}{y^2} = 4x - 2 + ce^{-2x}$$
$$\Rightarrow y^2 = \frac{1}{4x - 2 + ce^{-2x}}$$

Substitute $y = \frac{1}{3}$ when $x = 0$:

$$\left(\frac{1}{3}\right)^2 = \frac{1}{4 \times 0 - 2 + ce^{-2\times 0}}$$
$$\Rightarrow \frac{1}{9} = \frac{1}{-2 + c}$$
$$\Rightarrow c = 11$$
$$\therefore y^2 = \frac{1}{4x - 2 + 11e^{-2x}}$$

42)

$$x^3e^y\frac{\mathrm{d}y}{\mathrm{d}x} = 6x^2e^x - 3x^2e^y$$
$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{6x^2e^x}{x^3e^y} - \frac{3x^2e^y}{x^3e^y}$$
$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{6e^x}{xe^y} - \frac{3}{x}$$

Let $u = e^y \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = e^y\frac{\mathrm{d}y}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{e^y}\frac{\mathrm{d}u}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x}$

$$\therefore \frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{6e^x}{xu} - \frac{3}{x}$$
$$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{6e^x}{x} - \frac{3u}{x}$$

Rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x} + \mathrm{P}(x)u = \mathrm{Q}(x)$:

$$\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{3u}{x} = \frac{6e^x}{x}$$

Integrating Factor

$$\begin{aligned} &= e^{\int \frac{3}{x}\mathrm{d}x} \\ &= e^{3\ln|x|} \\ &= e^{\ln|x^3|} \\ &= |x^3| \end{aligned}$$

Let Integrating Factor $= x^3$

$$\begin{aligned} \frac{\mathrm{d}u}{\mathrm{d}x} + \frac{3u}{x} &= \frac{6e^x}{x} \\ \Rightarrow x^3\frac{\mathrm{d}u}{\mathrm{d}x} + 3x^2u &= 6x^2e^x \\ \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(x^3u) &= 6x^2e^x \end{aligned}$$

Use Integration by Parts:

$$\begin{aligned} u &= 6x^2 & \frac{\mathrm{d}v}{\mathrm{d}x} &= e^x \\ \frac{\mathrm{d}u}{\mathrm{d}x} &= 12x & v &= e^x \end{aligned}$$

$$\therefore \int 6x^2e^x\mathrm{d}x = 6x^2e^x - \int 12xe^x\mathrm{d}x$$

Use Integration by Parts again:

$$\begin{aligned} u &= 12x & \frac{\mathrm{d}v}{\mathrm{d}x} &= e^x \\ \frac{\mathrm{d}u}{\mathrm{d}x} &= 12 & v &= e^x \end{aligned}$$

$$\begin{aligned} \therefore \int 6x^2e^x\mathrm{d}x &= 6x^2e^x - \left(12xe^x - \int 12e^x\mathrm{d}x\right) \\ &= 6x^2e^x - 12xe^x + \int 12e^x\mathrm{d}x \\ &= 6x^2e^x - 12xe^x + 12e^x + c \\ \therefore x^3u &= 6x^2e^x - 12xe^x + 12e^x + c \\ \Rightarrow e^y &= \frac{6e^x}{x} - \frac{12e^x}{x^2} + \frac{12e^x}{x^3} + \frac{c}{x^3} \end{aligned}$$

$$\Rightarrow y = \ln\left(\frac{6x^2e^x - 12xe^x + 12e^x + c}{x^3}\right)$$

Substitute $y = \ln 6$ when $x = 1$:

$$\begin{aligned}
\ln 6 &= \ln\left(\frac{6 \times 1^2 \times e^1 - 12 \times 1 \times e^1 + 12 \times e^1 + c}{1^3}\right) \\
\Rightarrow \ln 6 &= \ln\left(\frac{6e - 12e + 12e + c}{1}\right) \\
\Rightarrow \ln 6 &= \ln(6e + c) \\
\Rightarrow 6 &= 6e + c \\
\Rightarrow c &= 6 - 6e \\
\therefore y &= \ln\left(\frac{6x^2e^x - 12xe^x + 12e^x + 6 - 6e}{x^3}\right)
\end{aligned}$$

43)

$$x\frac{\mathrm{d}y}{\mathrm{d}x} = y(\ln y + x)$$

Let $u = \ln y \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = y\frac{\mathrm{d}u}{\mathrm{d}x}$
$u = \ln y \Rightarrow y = e^u$

$$\begin{aligned}
\therefore xy\frac{\mathrm{d}u}{\mathrm{d}x} &= y(u + x) \\
\Rightarrow xe^u\frac{\mathrm{d}u}{\mathrm{d}x} &= e^u(u + x)
\end{aligned}$$

Rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x} + \mathrm{P}(x)u = \mathrm{Q}(x)$:

$$\begin{aligned}
\Rightarrow xe^u\frac{\mathrm{d}u}{\mathrm{d}x} &= ue^u + xe^u \\
\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} &= \frac{ue^u}{xe^u} + \frac{xe^u}{xe^u} \\
\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} &= \frac{u}{x} + 1 \\
\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} - \frac{u}{x} &= 1
\end{aligned}$$

Integrating Factor

$$\begin{aligned}
&= e^{\int \frac{-1}{x}\mathrm{d}x} \\
&= e^{-\ln|x|} \\
&= e^{\ln|x^{-1}|} \\
&= |x^{-1}|
\end{aligned}$$

Let Integrating Factor $= x^{-1} = \frac{1}{x}$

$$\begin{aligned}
\frac{\mathrm{d}u}{\mathrm{d}x} - \frac{u}{x} &= 1 \\
\Rightarrow \frac{1}{x}\frac{\mathrm{d}u}{\mathrm{d}x} - \frac{u}{x^2} &= \frac{1}{x} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u}{x}\right) &= \frac{1}{x} \\
\Rightarrow \frac{u}{x} &= \ln x + c \\
\Rightarrow u &= x\ln x + cx \\
\Rightarrow \ln y &= x\ln x + cx \\
\Rightarrow y &= e^{x\ln x + cx}
\end{aligned}$$

44)

$$\frac{\mathrm{d}y}{\mathrm{d}x} = e^x \cos^2 y + \sin y \cos y$$

Let $u = \tan y \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \sec^2 y \frac{\mathrm{d}y}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \cos^2 y \frac{\mathrm{d}u}{\mathrm{d}x}$

$$\begin{aligned}
\therefore \cos^2 y \frac{\mathrm{d}u}{\mathrm{d}x} &= e^x \cos^2 y + \sin y \cos y \\
\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} &= e^x + \frac{\sin y \cos y}{\cos^2 y} \\
\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} &= e^x + \tan y \\
\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} &= e^x + u
\end{aligned}$$

Rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x} + \mathrm{P}(x)u = \mathrm{Q}(x)$:

$$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} - u = e^x$$

Integrating Factor

$$\begin{aligned}
&= e^{\int -1 \mathrm{d}x} \\
&= e^{-x}
\end{aligned}$$

$$\begin{aligned}
\frac{\mathrm{d}u}{\mathrm{d}x} - u &= e^x \\
\Rightarrow e^{-x}\frac{\mathrm{d}u}{\mathrm{d}x} - ue^{-x} &= e^x e^{-x} \\
\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(ue^{-x}) &= 1 \\
\Rightarrow ue^{-x} &= x + c \\
\Rightarrow u &= (x + c)e^x \\
\Rightarrow \tan y &= (x + c)e^x \\
\Rightarrow y &= \arctan((x + c)e^x)
\end{aligned}$$

45 a)

$$p\frac{\mathrm{d}y}{\mathrm{d}x} + qy = ry^{1-n}$$

Let $u = y^n \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = ny^{n-1}\frac{\mathrm{d}y}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{ny^{n-1}}\frac{\mathrm{d}u}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y^{1-n}}{n}\frac{\mathrm{d}u}{\mathrm{d}x}$

$$\begin{aligned}
\therefore p \times \frac{y^{1-n}}{n}\frac{\mathrm{d}u}{\mathrm{d}x} + qy &= ry^{1-n} \\
\Rightarrow py^{1-n}\frac{\mathrm{d}u}{\mathrm{d}x} + qny &= rny^{1-n}
\end{aligned}$$

$$\Rightarrow p\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{qny}{y^{1-n}} = \frac{rny^{1-n}}{y^{1-n}}$$
$$\Rightarrow p\frac{\mathrm{d}u}{\mathrm{d}x} + qny^n = rn$$
$$\Rightarrow p\frac{\mathrm{d}u}{\mathrm{d}x} + qnu = rn$$

Rearrange into the form $\frac{\mathrm{d}u}{\mathrm{d}x} + \mathrm{P}(x)u = \mathrm{Q}(x)$:

$$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} + \frac{qn}{p}u = \frac{rn}{p}$$

Integrating Factor

$$= e^{\int \frac{qn}{p}\mathrm{d}x}$$
$$= e^{\frac{qn}{p}x}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{qn}{p}u = \frac{rn}{p}$$
$$\Rightarrow e^{\frac{qn}{p}x}\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{qn}{p}e^{\frac{qn}{p}x}u = \frac{rn}{p}e^{\frac{qn}{p}x}$$
$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(ue^{\frac{qn}{p}x}) = \frac{rn}{p}e^{\frac{qn}{p}x}$$
$$\Rightarrow ue^{\frac{qn}{p}x} = \frac{rn}{p} \times \frac{p}{qn}e^{\frac{qn}{p}x} + c$$
$$\Rightarrow ue^{\frac{qn}{p}x} = \frac{r}{q}e^{\frac{qn}{p}x} + c$$
$$\Rightarrow u = \frac{r}{q} + ce^{-\frac{qn}{p}x}$$
$$\Rightarrow y^n = \frac{r}{q} + ce^{-\frac{qn}{p}x}$$
$$\Rightarrow y = \sqrt[n]{\frac{r}{q} + ce^{-\frac{qn}{p}x}}$$

45 b)
As $x \to \infty$, $e^{-\frac{qn}{p}x} \to 0$ so $y \to \sqrt[n]{\frac{r}{q}}$